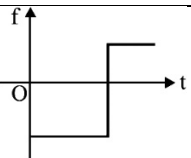
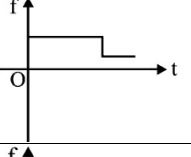
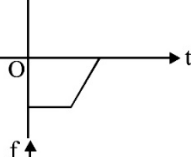
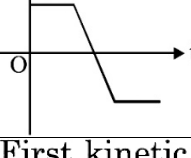
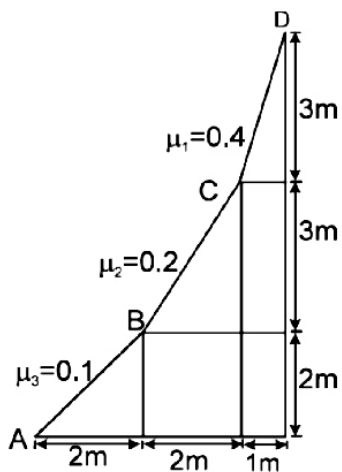
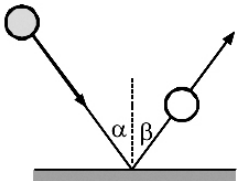
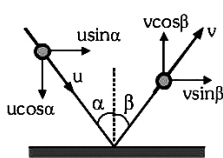
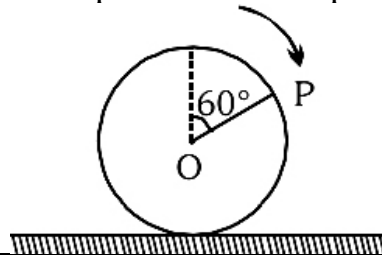
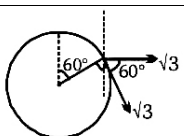
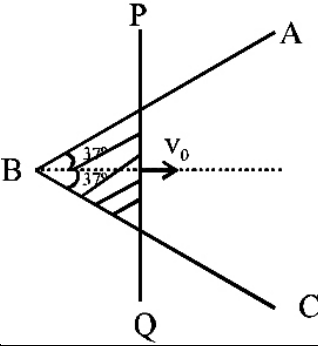
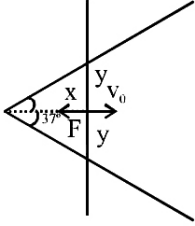
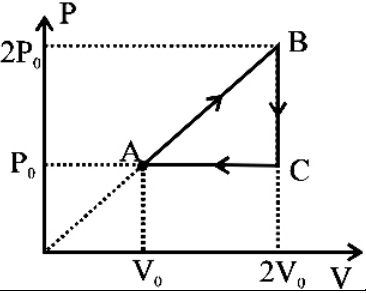
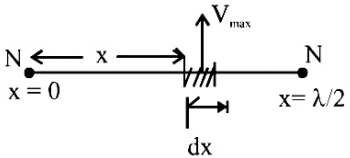
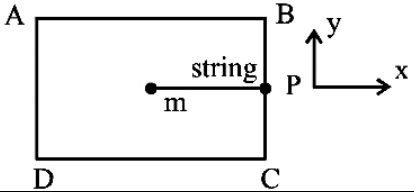
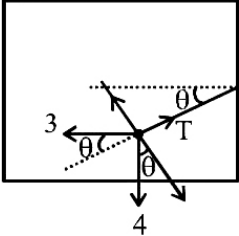


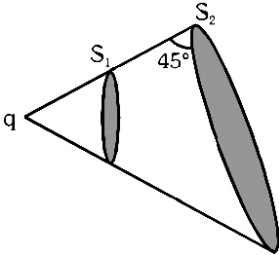
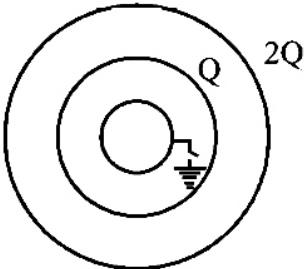
1.	A block is projected up along the line of greatest slope of an inclined plane of angle of inclination $\theta (< \phi)$ where ϕ = angle of friction. Which of the following friction-time graph is correct?	A
		
		
		
		
Sol.	First kinetic friction acts down the incline then static friction acts up the incline.	
2.	When a cyclist takes turn on a level road with constant speed.	C
	Net friction on him may be along tangential direction.	
	Net friction on him may be along outward radial direction.	
	Net reaction from ground should pass through center of mass of cyclist.	
	Net reaction from ground may not pass from center of mass of cyclist.	
Ans.	C	
3.	As shown in figure a body of mass 1 kg is shifted from A to D on inclined planes by applying a force slowly such that the block is always in contact with the plane surfaces. Neglecting the jerk experienced at points C and B, total work done by the force is :	A
		
	90 J	
	56 J	
	180 J	
	0 J	

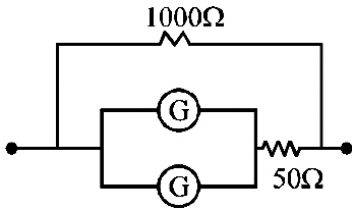
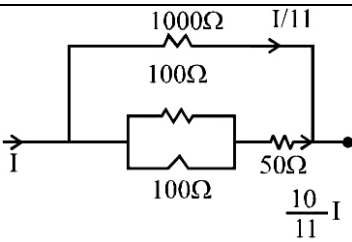
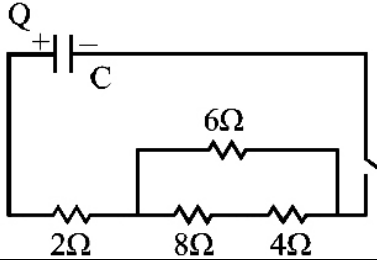
Sol.	$W = 1 \times 10 \times 8 + 0.1 \times 10 \times 2$ $+ 0.2 \times 10 \times 2 + 0.4 \times 10 \times 1$ $= 80 + 2 + 4 + 4$ $= 90 \text{ J}$	
4.	A ball is of mass m, strikes a smooth ground at angle α as shown in figure and is reflected at angle β. The coefficient of restitution will be 	A
	$\frac{\tan \alpha}{\tan \beta}$	
	$\frac{\cos \alpha}{\cos \beta}$	
	$\frac{\sin \alpha}{\sin \beta}$	
	$\frac{\tan \beta}{\tan \alpha}$	
Sol.	$v \sin \beta = u \sin \alpha$ and $e = \frac{v \cos \beta}{u \cos \alpha} \Rightarrow e = \frac{\tan \alpha}{\tan \beta}$ 	
5.	A wheel of radius $R = 0.1 \text{ m}$ is rolling without slipping on a horizontal surface as shown in the figure. Centre of the wheel moves with a constant speed $\sqrt{3} \text{ m/s}$. The speed of the point P with respect to ground is :- 	C
	$2\sqrt{3} \text{ m/s}$	
	Zero	
	3 m/s	
	$\sqrt{3} \text{ m/s}$	
Sol.	 $\sqrt{(\sqrt{3})^2 + (\sqrt{3})^2 + 2(\sqrt{3})(\sqrt{3})\cos 60^\circ} = 3 \text{ m/s}$	
6.	ABC is a smooth horizontal wire frame. PQ is a wire of mass m, which can slide on frame. There is a film of a liquid of surface tension T covering the triangular part. Wire PQ is given a velocity v_0 when it is at a distance x_0 from B. Find v_0 if wire PQ is just able to reach upto a distance $2x_0$ from B.	C

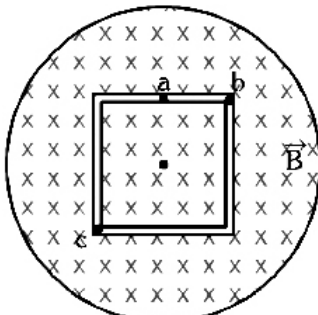
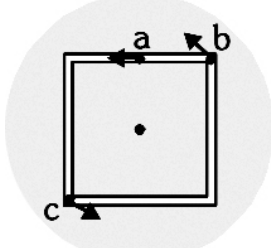
			
	$x_0 \sqrt{\frac{T}{m}}$		
	$2x_0 \sqrt{\frac{T}{m}}$		
	$3x_0 \sqrt{\frac{T}{m}}$		
	$4x_0 \sqrt{\frac{T}{m}}$		
Sol.	$\frac{y}{x} = \tan 37^\circ = \frac{3}{4}$ $\Rightarrow y = \frac{3x}{4}$ $F = T \cdot 2(2y)$ $= 4T \times \frac{3x}{4}$ $F = 3Tx$ (Along -ve x)  from work energy theorem $\int_{x_0}^{2x_0} (-3Tx) dx = 0 - \frac{1}{2}mv_0^2$ $\Rightarrow -3T \cdot \left(\frac{x^2}{2} \right)_{x_0}^{2x_0} = -\frac{mv_0^2}{2}$ $\Rightarrow 9x_0^2 T = mv_0^2 \Rightarrow v_0 = 3x_0 \sqrt{\frac{T}{m}}$		
7.	A mono-atomic gas undergoes a cyclic process as shown. Find efficiency. 		D
	$\frac{1}{6}$		
	$\frac{1}{7}$		
	$\frac{1}{11}$		
	$\frac{1}{12}$		

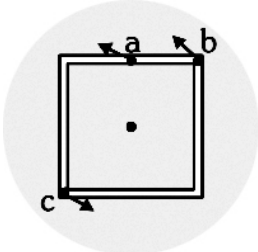
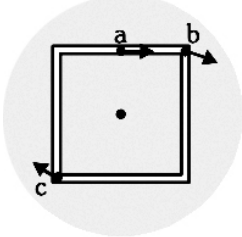
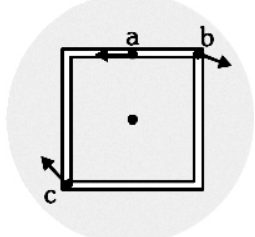
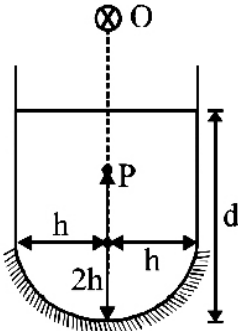
Sol.	$\text{AB, } w_1 = \text{Area} = \frac{3}{2} P_0 V_0$ $\Delta U_1 = \frac{3}{2} nR(T_B - T_A) = \frac{3}{2} (P_B V_B - P_A V_A)$ $= \frac{9}{2} P_0 V_0$ $\therefore Q_1 = \Delta U_1 + W_1 = 6P_0 V_0$	$\text{BC } W_2 = 0 \text{ and } Q_2 = -ve$ $\text{CA } W_3 = -P_0 V_0 (\text{Area}) \text{ and } Q_3 = -ve$ $\therefore \eta = \frac{w_1 + w_2 + w_3}{Q(+ve)} = \frac{\frac{3}{2} P_0 V_0 + 0 - P_0 V_0}{6P_0 V_0}$ $= \frac{1}{12}$	
8.	Equation of a standing wave on a string of linear mass density m is given by $y = A \sin kx \sin \omega t$. Maximum particle velocity at antinode is v_0 . If λ be the wavelength of superposing waves then find total maximum kinetic energy of all particles within two consecutive nodes.		A
	$\frac{\mu \lambda v_0^2}{8}$		
	$\frac{\mu \lambda v_0^2}{4}$		
	$\frac{\mu \lambda v_0^2}{2}$		
	$\mu \lambda v_0^2$		
Sol.	$y = A \sin kx \sin \omega t$  For an element at position x, $v = \frac{\partial y}{\partial t} = A \sin kx \cos \omega t \cdot \omega$ $v = (\omega A) \sin kx \cos \omega t$	$\therefore v_{\max} = (\omega A) \sin kx = v_0 \sin kx$ At antinode $\sin kx = 1 \therefore v_0 = \omega A$ Now $dk_{\max} = \frac{1}{2} dm \cdot v_{\max}^2$ $= \frac{1}{2} \cdot \mu dx \cdot v_0^2 \sin^2 kx$ $\therefore k_{\max} = \frac{\mu v_0^2}{2} \int_0^{\lambda/2} \sin^2 kx dx = \frac{\mu v_0^2 \lambda}{8}$	
9.	ABCD is a smooth plank placed in horizontal xy-plane. A small body of mass m is connected to a string which is connected to point P of plank. Initially plank and body both are at rest. Now an acceleration of $3\hat{i} + 4\hat{j} \text{ m/s}^2$ is given to plank. Find angle made by string with its initial direction when net force on body is along the string as seen from the frame of the plank. 		B
	37°		
	53°		
	$\tan^{-1} \frac{4}{5}$		
	$\tan^{-1} \frac{3}{5}$		

Sol.	 $3 \sin \theta = 4 \cos \theta$ $\Rightarrow \tan \theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$	
10.	Density of a solid sphere of radius R is $\rho = \rho_0 \left(1 + \frac{r}{R}\right)$ at a distance r from the centre. Find magnitude of intensity of gravitational $r = \frac{R}{2}$ from centre.	C
	$\left(\frac{4\pi}{3}\right) \rho_0 GR$	
	$\left(\frac{9\pi}{8}\right) \rho_0 GR$	
	$\left(\frac{11\pi}{12}\right) \rho_0 GR$	
	$\left(\frac{13\pi}{14}\right) \rho_0 GR$	
Sol.	From Gauss law $E \cdot 4\pi r^2 = 4\pi G \int \rho dv$ $\Rightarrow E = \frac{G}{r^2} \int_0^r \rho_0 \left(1 + \frac{r}{R}\right) \cdot 4\pi r^2 dr$ $= \frac{G}{r^2} \cdot 4\pi \rho_0 \int_0^r \left(r^2 + \frac{r^3}{R}\right) dr$ $= \frac{4\pi \rho_0 G}{r^2} \cdot \left(\frac{r^3}{3} + \frac{r^4}{4R}\right) = 4\pi \rho G \left(\frac{r}{3} + \frac{r^2}{4R}\right)$ $\therefore \text{ at } r = \frac{R}{2}$ $E = 4\pi \rho_0 G \left(\frac{R}{6} + \frac{R}{16}\right)$ $E = \left(\frac{11\pi}{12}\right) \rho_0 GR$	
11.	When a body is placed in surroundings at a constant temperature of 20°C , and heated by a 10-W heater, its temperature remains constant at 40°C . If the temperature of the body is now raised from 20°C to 80°C in 5 minutes at a uniform rate, the total heat it will lose to the surroundings will be- (Assume Newton's law of cooling is valid)	C
	3000 J	
	3600 J	
	4500 J	
	5400 J	

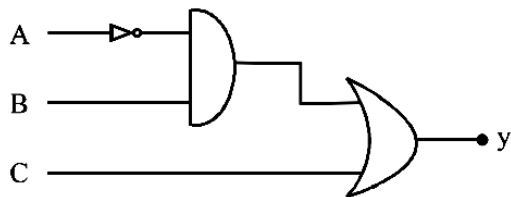
Sol.	Let θ = the temperature of the body, θ_0 = the temperature of the surroundings $= 20^\circ\text{C}$ (given). Rate of loss of heat $= \frac{dQ}{dt} = k(\theta - \theta_0)$ For $\theta = 40^\circ\text{C}$, $\frac{dQ}{dt} = 10 = k(40 - 20) \Rightarrow k = \frac{1}{2}$ When the temperature of the body is raised uniformly, its temperature is given by $\theta = 20 + \left(\frac{80 - 20}{300\text{s}}\right)t = 20 + \frac{t}{5}$, where t = time in seconds $\therefore \frac{dQ}{dt} = k(\theta - \theta_0) = \frac{1}{2} \left(\frac{t}{5}\right) = \frac{t}{10}$ $\therefore Q = \int_0^{300} \frac{t}{10} dt = \frac{1}{20} (300)^2 \text{ J} = 4500 \text{ J}$	
12.	In the given figure flux through surface S_1 is ϕ_1 & through S_2 is ϕ_2. Which is correct ? 	A
	$\phi_1 = \phi_2$	
	$\phi_1 > \phi_2$	
	$\phi_1 < \phi_2$	
	None of these	
Sol.	Both cuts equal number of electric lines	
13.	Three concentric spherical conducting shells having radii R, $2R$ and $3R$ are shown in figure. Charge on inner most sphere after it is earthed is : 	B
	$-\frac{2Q}{3}$	
	$-\frac{7}{6}Q$	
	$-\frac{Q}{6}$	
	$-\frac{5}{6}Q$	

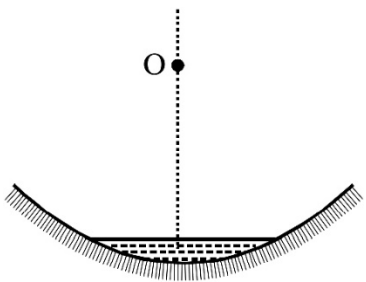
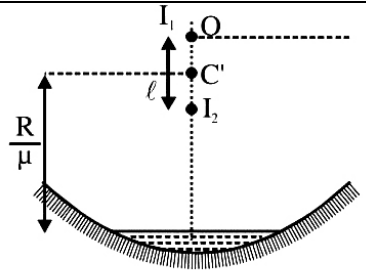
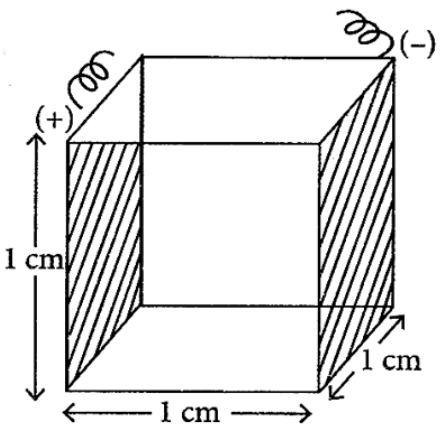
Sol.	$\frac{kq}{R} + \frac{kQ}{2R} + \frac{k(2Q)}{3R} = 0$ $q = -\frac{7Q}{6}$	
14.	Given circuit represents an ammeter. Resistance of each Galvanometer coil is 100Ω and shows maximum deflection at 10mA current. Range of ammeter is 	A
	22 mA	
	26 mA	
	28 mA	
	24 mA	
Sol.	 $I_g = \frac{5}{11} I$ $I = \frac{11}{5} I_g$ $I_{\max} = \frac{11}{5} \cdot 10 = 22\text{mA}$	
15.	Amount of heat produced in 8W resistor after key is closed is : 	C
	$\frac{4Q^2}{27c}$	
	$\frac{4Q^2}{27c}$	
	$\frac{2Q^2}{27c}$	
	$\frac{10Q^2}{27c}$	

Sol.	Equivalent resistance = 6Ω if current is I, current through $8\Omega = \frac{I}{3}$. now, $P_{\text{total}} = 6I^2 \quad \dots (1)$ $P = \left(\frac{I}{3}\right)^2 \cdot 8 = \frac{8}{9}I^2 = \text{Power through } 8\Omega.$ Now, heat developes = $\frac{P}{P_{\text{total}}} \left(\frac{Q^2}{2c}\right) = \frac{2Q^2}{27c}$	
16.	A 100 turn closely wound circular coil of radius 10 cm carries a current of 3.2 A. The coil is placed in a vertical plane and is free to rotate about a horizontal axis which coincides with its diameter. A uniform magnetic field of 2T in the horizontal direction exists such that initially the axis of the coil is in the direction of the field. What is the approximate angular speed aquired by the coil when it has rotated by 90°? The moment of inertia of the coil about given diameter is 0.1 kg m^2. 35 rad s^{-1} 25 rad s^{-1} 20 rad s^{-1} 30 rad s^{-1}	C
Sol.	$I \left(\omega \frac{d\omega}{d\theta} \right) = MB \sin \theta$ $M = NIA$	
17.	A square conducting loop is placed in the time varying magnetic field $\left(\frac{dB}{dt} = -ve \text{ constant}\right)$. The centre of square coincides with axis of cylindrical region of magnetic field. The directions of induced electric field at point a, b and c.  The diagram shows a square conducting loop centered within a circular region representing a magnetic field. The magnetic field is directed into the page, indicated by 'x' marks. Points 'a', 'b', and 'c' are marked on the top, right, and bottom sides of the square loop, respectively.	C
	 The diagram shows the square loop with arrows indicating the direction of the induced electric field at points 'a', 'b', and 'c'. At point 'a' (top), the arrow points to the left. At point 'b' (right), the arrow points upwards. At point 'c' (bottom), the arrow points to the right.	

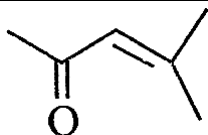
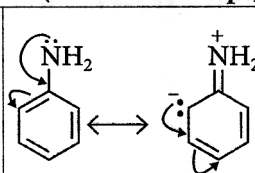
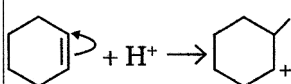
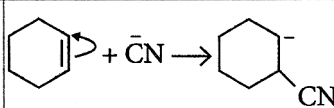
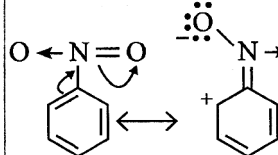
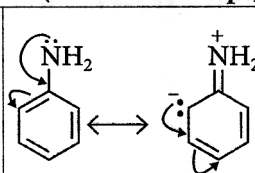
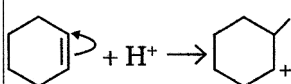
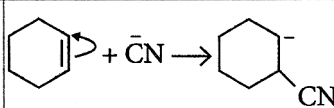
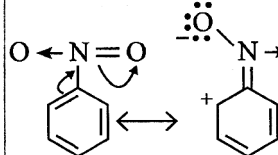
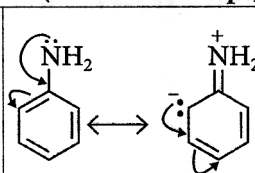
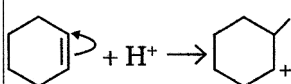
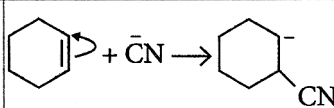
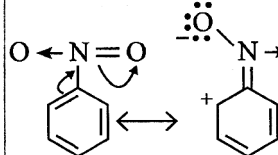
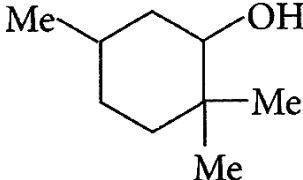
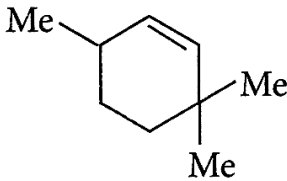
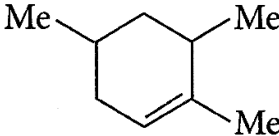
		
		
		
Sol.	Since flux decreasing field line will be such that it will increasing the 'B'.	
18.	A concave mirror of radius h is placed at the bottom of a tank containing a liquid of refractive index μ upto a depth d. An object P is placed at height $2h$ above the bottom of the mirror. Outside the liquid, an observer O views the object and its image in the mirror. The apparent distance between these two will be:- [$d > 2h$] 	B
	Zero	
	$\frac{4h}{3\mu}$	
	$h\left(1 - \frac{1}{\mu}\right)$	
	$\frac{2h}{\mu - 1}$	

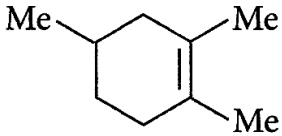
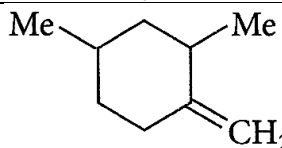
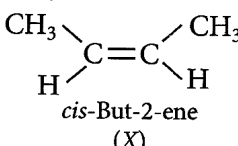
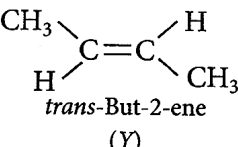
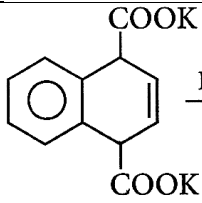
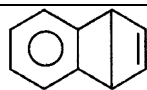
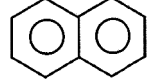
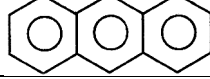
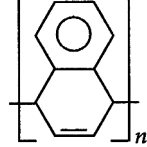
Sol.	$\frac{1}{v} - \frac{1}{2h} = -\frac{2}{h}$ $v = \frac{-2h}{3}$ distance of its image from observer $= \frac{d - \frac{2h}{3}}{\mu}$ apparent distance between these two $= \frac{d - \frac{2h}{3}}{\mu} - \frac{d - 2h}{\mu} = \frac{4h}{3\mu}$	
19.	Two polaroids are crossed to each other. When both of them are rotated through 30° in opposite sense, then what percentage of the incident unpolarised light will be transmitted by the polaroids ?	A
	37.50	
	35.5	
	39	
	41	
Sol.	Initially the polaroids are crossed to each other, that is the angle between their polarising directions is 90°. When both are rotated through 30° opposite sense, then the angle between their polarising directions will become 30°. Let the intensity of the incident unpolarised light = I_0 Then the intensity of light emerging from the first polaroid is $I_1 = \frac{1}{2}I_0$ This light is plane polarised and passes through the second polaroid.	The intensity of light emerging from the second polaroid is $I_2 = I_1 \cos^2 \theta$ θ = the angle between the polarising directions of the two polaroids. $I_1 = \frac{1}{2}I_0$ and $\theta = 30^\circ$ so $I_2 = I_1 \cos^2 30^\circ = \frac{1}{2}I_0 \cos^2 30^\circ$ $\Rightarrow \frac{I_2}{I_0} = \frac{3}{8}$ $\therefore$ transmission percentage $= \frac{I_2}{I_0} \times 100 = \frac{3}{8} \times 100 = 37.5\%$
20.	In a YDSE experiment $\lambda = 540$ nm, $D = 1$ m, $d = 1$ mm. A thin film is pasted on upper slit and the central maxima shifts to the point just in front of the upper slit. What is the path difference at the centre of the screen ?	C
	540 nm	
	270 nm	
	500 nm	
	810 nm	
Sol.	$\frac{d}{2} = (\mu - 1)t \frac{D}{d}$ $(\mu - 1)t = \frac{d^2}{2D}$ $\Delta x = (\mu - 1)t = \frac{d^2}{2D}$	
21.	A simple pendulum which has spherical bob is oscillating in container which is filled with viscous fluid (viscosity coefficient = η). Find angular frequency of damped oscillation of simple pendulum. Given radius of the bob is 50 cm, length of simple pendulum is 2 m, density of the bob is 3 kg/m^3 and $\eta = \frac{1}{3} \text{ N}$	2

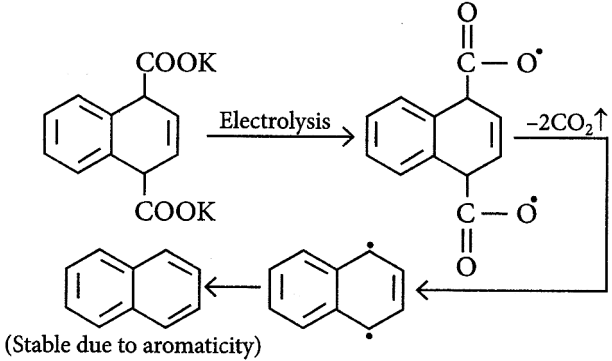
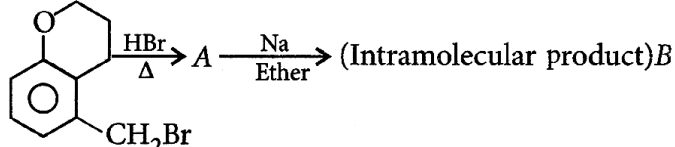
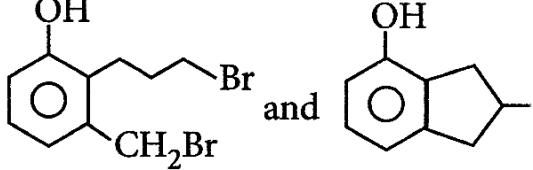
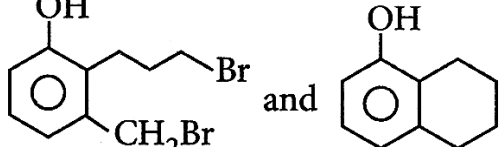
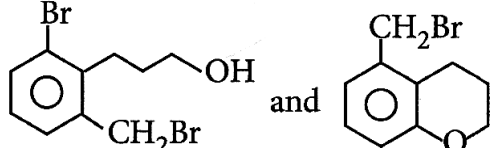
	sm^{-2} .																																														
Sol.	$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2} = \sqrt{\frac{g}{l} - \left(\frac{6\pi\eta r}{2m}\right)^2}$																																														
22.	The current required to be passed in a solenoid of length 10 cm and number of turns 100, so that the magnet gets demagnetized when inside the solenoid is 3A. The coercivity of a small magnet where the ferromagnet gets demagnetized is $n \times 10^2 \text{ A m}^{-1}$. Find the value of n.	30																																													
Sol.	$B = \mu_0 H = \mu_0 n i$ $n \times 10^2 = \frac{100}{10 \times 10^{-2}} \times 3$ $n = 30$																																														
23.	In n cases of input output is $Y = 1$ in the given diagram. Then find the value of n.  <table border="1"><thead><tr><th></th><th>A</th><th>B</th><th>C</th><th>Y</th></tr></thead><tbody><tr><td>1</td><td>0</td><td>0</td><td>0</td><td></td></tr><tr><td>2</td><td>0</td><td>0</td><td>1</td><td></td></tr><tr><td>3</td><td>0</td><td>1</td><td>0</td><td></td></tr><tr><td>4</td><td>0</td><td>1</td><td>1</td><td></td></tr><tr><td>5</td><td>1</td><td>0</td><td>0</td><td></td></tr><tr><td>6</td><td>1</td><td>0</td><td>1</td><td></td></tr><tr><td>7</td><td>1</td><td>1</td><td>0</td><td></td></tr><tr><td>8</td><td>1</td><td>1</td><td>1</td><td></td></tr></tbody></table>		A	B	C	Y	1	0	0	0		2	0	0	1		3	0	1	0		4	0	1	1		5	1	0	0		6	1	0	1		7	1	1	0		8	1	1	1		5
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Ans.	5																																														
24.	A mixture of 250 gm of water and 200 gm of ice are at 0°C is kept in a calorimeter having water equivalent of 50 gm. If 200 gm of steam at 100°C is passed through the mixture, then find the final temperature of mixture in $^\circ\text{C}$: ($L_f = 80 \text{ cal/gm}$, $L_v = 540 \text{ cal/gm}$, $S_w = 1 \text{ cal/gm-}^\circ\text{C}$, $S_{ice} = 0.5 \text{ cal/gm-}^\circ\text{C}$)	100																																													
Sol.	Amount of heat required to take ice + water mixture at 100°C. $((250+50) \times 1 \times 100) + (200 \times 80 + 200 \times 1 \times 100)$ $= 66,000 \text{ cal.}$ But the amount of energy released by steam while coming to 100°C water is $200 \times 540 = 108,000.$ Which is more than heat required $\therefore$ final temperature is 100°C and some part of steam will remain as it is.																																														
25.	The concave spherical mirror of radius of $R = 9 \text{ cm}$ forms the image of a source which coincides with a source. When a little liquid is poured in the mirror, a second image is formed between the source and the mirror at a distance $\ell = 4 \text{ cm}$ from the source. Find the refractive index of the liquid.	7																																													

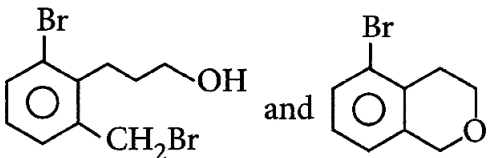
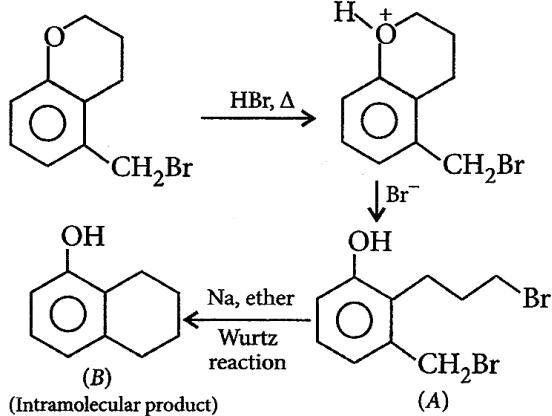
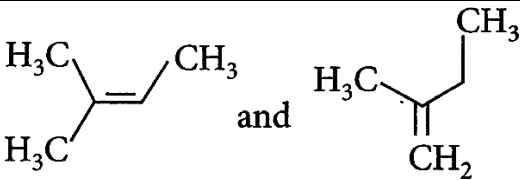
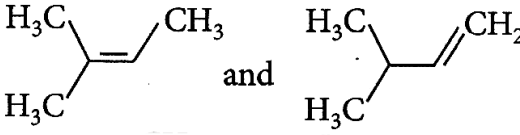
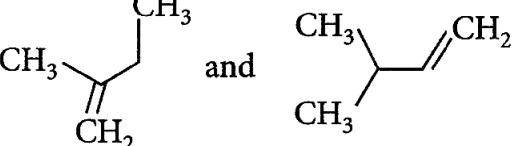
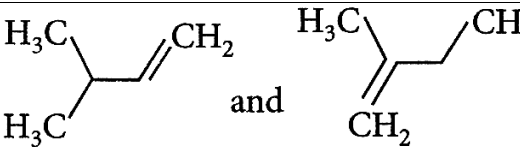
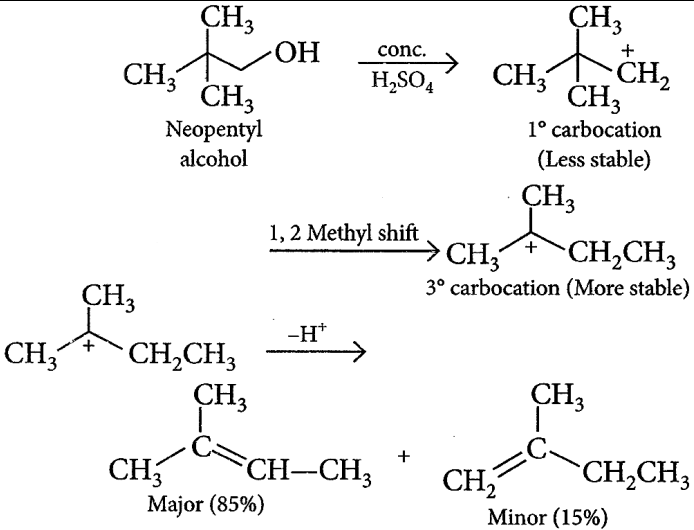
	Write '5μ' in OMR Sheet.	
		
Sol.	 On pouring water the new center of curvature is at R/μ. $\frac{1}{v} + \frac{1}{-R} = \frac{2\mu}{-R}$ $\therefore v = \frac{R}{1-2\mu} \text{ for } I_2; v = -5$ $\mu = 1.4$	
26.	A conductivity cell with two electrodes (dark side) are half filled with infinitely dilute aqueous solution of a weak electrolyte. If volume is doubled by adding more water at constant temperature, the molar conductivity of the cell will 	C
	Depend upon type of electrolyte	
	Increase sharply	
	Remain same or cannot be measured accurately	
	Decrease sharply	
Sol.	Molar conductivity of weak electrolytes is maximum at infinite dilution, further dilution does not affect it.	
27.	The bond dissociation energy of B – F in BF_3 is 646 kJ mol^{-1} whereas that of C – F in CF_4 is 515 kJ mol^{-1}. The correct reason for higher B – F bond dissociation energy as compared to that of C–F is smaller size of B-atom as compared to that of C-atom stronger σ bond between B and F in BF_3 as compared to that between C and F	C

	in CF ₄																									
	significant pπ – pπ interaction between B and F in BF ₃ whereas there is no possibility of such interaction between C and F in CF ₄ .																									
	lower degree of pπ – pπ interaction between B and F in BF ₃ than that between C and F in CF ₄																									
Sol.	In BF ₃ , B is sp ² hybridised and has a vacant 2p-orbital which overlaps laterally with a filled 2p-orbital of F forming strong pπ-pπ bond. However in CF ₄ , C does not have any vacant p-orbitals to undergo π-bonding. Thus B.E. _{B – F} > B.E. _{C – F} .																									
28.	Match List– I with List –II. <table border="1"><thead><tr><th colspan="2">List - I (Name of oxo acid)</th><th colspan="2">List - II (Oxidation state of 'P')</th></tr></thead><tbody><tr><td>(A)</td><td>Hypophosphorous acid</td><td>(i)</td><td>+5</td></tr><tr><td>(B)</td><td>Orthophosphoric acid</td><td>(ii)</td><td>+4</td></tr><tr><td>(C)</td><td>Hypophosphoric acid</td><td>(iii)</td><td>+3</td></tr><tr><td>(D)</td><td>Orthophosphorous acid</td><td>(iv)</td><td>+2</td></tr><tr><td></td><td></td><td>(v)</td><td>+1</td></tr></tbody></table> Choose the correct answer from the options given below.	List - I (Name of oxo acid)		List - II (Oxidation state of 'P')		(A)	Hypophosphorous acid	(i)	+5	(B)	Orthophosphoric acid	(ii)	+4	(C)	Hypophosphoric acid	(iii)	+3	(D)	Orthophosphorous acid	(iv)	+2			(v)	+1	D
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Sol.	<table border="1"><thead><tr><th>Acid</th><th>Formula</th><th>Oxid. state of P</th></tr></thead><tbody><tr><td>Hypophosphorous acid</td><td>H₃PO₂</td><td>+1</td></tr><tr><td>Orthophosphoric acid</td><td>H₃PO₄</td><td>+5</td></tr><tr><td>Hypophosphoric acid</td><td>H₄P₂O₆</td><td>+4</td></tr><tr><td>Orthophosphorous acid</td><td>H₃PO₃</td><td>+3</td></tr></tbody></table>	Acid	Formula	Oxid. state of P	Hypophosphorous acid	H ₃ PO ₂	+1	Orthophosphoric acid	H ₃ PO ₄	+5	Hypophosphoric acid	H ₄ P ₂ O ₆	+4	Orthophosphorous acid	H ₃ PO ₃	+3										
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29.	Strong reducing and oxidizing agents among the following, respectively, are	B																								
	Ce ⁴⁺ and Eu ⁺²																									
	Eu ⁺² and Ce ⁴⁺																									
	Ce ⁴⁺ and Ce ⁴⁺																									
	Ce ⁴⁺ and Tb ⁴⁺																									
Sol.	The stable oxidation state of lanthanoids is +3.Ce ⁴⁺ tends to accept an electron to change to +3 state. Hence, it acts as a good oxidising agent. Eu ⁺² has a strong tendency to lose an electron to attain the more stable +3 oxidation state of lanthanoids. Hence, it is a strong reducing agent.																									
30.	After understanding the assertion and reason, choose the correct option. Assertion : In the bonding molecular orbital (MO) of H ₂ , electron density is increased between the nuclei. Reason : The bonding MO is ψ _A + ψ _B , which shows destructive interference of the combining electron waves.	C																								
	Assertion and reason are correct and reason is the correct explanation for the assertion.																									
	Assertion and reason are correct, but reason is not the correct explanation for the assertion.																									

	Assertion is correct, reason is incorrect.																					
	Assertion is incorrect, reason is correct																					
Sol.	Bonding molecular orbital involves constructive interference.																					
31.	Mesityl oxide is a common name of	A																				
	4-methylpent-3-en-2-one																					
	2, 4-dimethylpentan-3-one																					
	2-methylcyclohexanone																					
	3-methylcyclohexane carbaldehyde																					
Sol.																						
32.	Match List I with List II. <table border="1" data-bbox="218 656 805 1314"><thead><tr><th colspan="2">List I (Mechanism steps)</th><th colspan="2">List II (Effect)</th></tr></thead><tbody><tr><td>(A)</td><td></td><td>(I)</td><td>-E effect</td></tr><tr><td>(B)</td><td></td><td>(II)</td><td>-R effect</td></tr><tr><td>(C)</td><td></td><td>(III)</td><td>+E effect</td></tr><tr><td>(D)</td><td></td><td>(IV)</td><td>+R effect</td></tr></tbody></table> Choose the correct answer from the options given below: (A) → (IV), (B) → (III) (C) → (I) (D) → (II) (A) → (III), (B) → (I) (C) → (II) (D) → (IV) (A) → (II), (B) → (IV) (C) → (III) (D) → (I) (A) → (I), (B) → (II) (C) → (IV) (D) → (III)	List I (Mechanism steps)		List II (Effect)		(A)		(I)	-E effect	(B)		(II)	-R effect	(C)		(III)	+E effect	(D)		(IV)	+R effect	A
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(B)		(II)	-R effect																			
(C)		(III)	+E effect																			
(D)		(IV)	+R effect																			
33.	The major product (P) of the given reaction is (where, Me is – CH ₃)  $\xrightarrow{H^+}$ P Major Product	C																				
																						
																						

		
		
34.	When 2-butyne is treated with H_2 /Lindlar's catalyst, compound X is produced as the major product and when treated with Na/liq. NH_3 it produces Y as the major product. Which of the following statements is correct?	B
	Y will have higher dipole moment and higher boiling point than X.	
	X will have higher dipole moment and higher boiling point than Y.	
	X will have lower dipole moment and lower boiling point than Y.	
	Y will have higher dipole moment and lower boiling point than X.	
Sol.	$CH_3-C\equiv C-CH_3 + H_2$ But-2-yne $\xrightarrow[\text{Lindlar's catalyst}]{\text{Pd/BaSO}_4 + \text{Quinoline}}$  <i>cis</i>-But-2-ene (X) $CH_3-C\equiv C-CH_3 \xrightarrow[\text{(ii) Ethanol, Birch reduction}]{\text{(i) Na/Liq. } NH_3, 196-200\text{ K}}$ But-2-yne  <i>trans</i>-But-2-ene (Y) Dipole moment of Y = 0 X has higher dipole moment and higher boiling point than Y.	
35.	 Electrolysis $\rightarrow$ A A is :	B
		
		
		
		

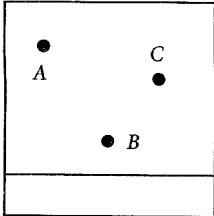
Sol	 (Stable due to aromaticity)	
36.	Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A): Haloalkanes react with KCN to form alkyl cyanides as a main product while with AgCN form isocyanide as the main product. Reaction (R): KCN and AgCN both are highly ionic compounds. In the light of the above statements, choose the most appropriate answer from the option given below.	D
	Both (A) and (R) are correct and (R) is the correct explanation of (A).	
	(A) is not correct but (R) is correct.	
	Both (A) and (R) are correct but (R) is not the correct explanation of (A).	
	(A) is correct but (R) is not correct.	
Sol.	KCN is predominantly ionic and provides cyanide ions in solution. Although both carbon and nitrogen atoms are in a position to donate electron pairs, the attack takes place mainly through carbon atom and not through nitrogen atom since C – C bond is more stable than C – N bond. However, AgCN is mainly covalent in nature and nitrogen is free to donate electron pair forming isocyanide as the main product.	
37.	In the following reaction sequence, structures of A and B, respectively will be 	B
		
		
		

		
Sol.	 (Intramolecular product) (A) (B)	
38.	When neopentyl alcohol is heated with an acid, it slowly converted into an 85 : 15 mixture of alkenes A and B, respectively. What are these alkenes?	A
		
		
		
		
Sol.	 Neopentyl alcohol 1° carbocation (Less stable) 1, 2 Methyl shift 3° carbocation (More stable) Major (85%) Minor (15%)	
39.	Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A): pK_a value of phenol is 10.0 while that of ethanol is 15.9. Reason (R): Ethanol is stronger acid than phenol.	B

	In the light of the above statements, choose the correct answer from the options given below :	
	(A) is false but (R) is true.	
	(A) is true but (R) is false	
	Both (A) and (R) are true but (R) is not the correct explanation of (A).	
	Both (A) and (R) are true and (R) is the correct explanation of (A).	
40.	In the following reaction : $ \begin{array}{c} \text{H}_3\text{C} \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{CH} \\ \\ \text{C}_6\text{H}_5 \end{array} + \text{O}_2 \longrightarrow \begin{array}{c} \text{H}_3\text{C} \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} - \text{O} - \text{O} - \text{H} \\ \\ \text{C}_6\text{H}_5 \end{array} \xrightarrow[\text{H}_2\text{O}]{\text{H}^+} \text{A and B} $ The compounds A and B respectively are	C
	$\text{C}_6\text{H}_5\text{CH}_3$, CH_3COOH	
	$\text{C}_6\text{H}_4(\text{OH})_2$, CH_3COCH_3	
	$\text{C}_6\text{H}_5\text{OH}$, CH_3COCH_3	
	$\text{C}_6\text{H}_3(\text{OH})_3$, CH_3COCH_3	
Sol.	$ \begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} - \text{O} - \text{O} - \text{H} \\ \\ \text{C}_6\text{H}_5 \end{array} \xrightarrow[\text{H}_2\text{O}]{\text{H}^+} \begin{array}{c} \text{OH} \\ \\ \text{C}_6\text{H}_5 \end{array} + \text{CH}_3 - \text{C}(=\text{O}) - \text{CH}_3 $	
41.	The major products A and B formed in the following reaction sequence are $ \text{C}_6\text{H}_5\text{NH}_2 \xrightarrow{\text{Ac}_2\text{O}} \text{A} \xrightarrow[\text{Room temperature}]{\text{Br}_2/\text{CH}_3\text{COOH}} \text{B} $	C
	A = $\text{C}_6\text{H}_4(\text{NH}_2)(\text{COCH}_3)$, B = $\text{C}_6\text{H}_3(\text{NH}_2)(\text{COCH}_3)(\text{Br})$	

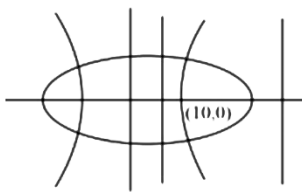
Sol.		
42.	L-isomer of a compound A ($C_4H_8O_4$) gives a positive test with $[Ag(NH_3)_2]^+$. Treatment of A with acetic anhydride yields triacetate derivative. Compound A produces an optically active compound (B) and an optically inactive compound (C) on treatment with bromine water and HNO_3 respectively. Compound (A) is	A

	$\begin{array}{c} \text{CHO} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{CH}_2\text{OH} \end{array}$																					
Sol.	$\begin{array}{c} \text{CHO} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{CH}_2\text{OH} \end{array}$ $\xrightarrow{[\text{Ag}(\text{NH}_3)_2]^+}$ $\begin{array}{c} \text{COO}^- \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{CH}_2\text{OH} \end{array}$ L-Erythrose When (A) is heated with acetic anhydride, acetylation occurs and —OH group is replaced by —OCOCH₃ thus, triacetate is formed. This confirms the presence of 3 —OH groups. $\begin{array}{c} \text{CHO} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{CH}_2\text{OH} \end{array}$ $\xrightarrow{\text{Br}_2/\text{H}_2\text{O}}$ $\begin{array}{c} \text{COOH} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{CH}_2\text{OH} \end{array}$ (A) (B) (Optically active) $\begin{array}{c} \text{CHO} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{CH}_2\text{OH} \end{array}$ $\xrightarrow{\text{conc. HNO}_3}$ $\begin{array}{c} \text{COOH} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{COOH} \end{array}$ (A) (C) (Optically inactive)																					
43.	Match List I with List II. <table><thead><tr><th colspan="2">List I (Complex)</th><th colspan="2">List II (Colour)</th></tr></thead><tbody><tr><td>A.</td><td>Mg(NH₄)PO₄</td><td>I.</td><td>Brown</td></tr><tr><td>B.</td><td>K₃[Co(NO₂)₆]</td><td>II.</td><td>White</td></tr><tr><td>C.</td><td>MnO(OH)₂</td><td>III.</td><td>Yellow</td></tr><tr><td>D.</td><td>Fe₄[Fe(CN)₆]₃</td><td>IV.</td><td>Blue</td></tr></tbody></table> Choose the correct answer from the options given below: A-II, B-III, C-IV, D-I A-II, B-III, C-I, D-IV A-III, B-IV, C-II, D-I A-II, B-IV, C-I, D-III	List I (Complex)		List II (Colour)		A.	Mg(NH ₄)PO ₄	I.	Brown	B.	K ₃ [Co(NO ₂) ₆]	II.	White	C.	MnO(OH) ₂	III.	Yellow	D.	Fe ₄ [Fe(CN) ₆] ₃	IV.	Blue	B
List I (Complex)		List II (Colour)																				
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D.	Fe ₄ [Fe(CN) ₆] ₃	IV.	Blue																			
Sol.	Fe₄[Fe(CN)₆]₃ has prussian blue colour. Mg(NH₄)PO₄ is white, K₃[Co(NO₂)₆] is yellow and MnO(OH)₂ is brown in colour.																					
44.	Match List-I with List-II. <table><thead><tr><th colspan="2">List-I (Metal ion)</th><th colspan="2">List-II (Group in qualitative analysis)</th></tr></thead><tbody><tr><td>(A)</td><td>Mn²⁺</td><td>(i)</td><td>Group-III</td></tr><tr><td>(B)</td><td>As³⁺</td><td>(ii)</td><td>Group-IIA</td></tr><tr><td>(C)</td><td>Cu²⁺</td><td>(iii)</td><td>Group-IV</td></tr><tr><td>(D)</td><td>Al³⁺</td><td>(iv)</td><td>Group-IIB</td></tr></tbody></table>	List-I (Metal ion)		List-II (Group in qualitative analysis)		(A)	Mn ²⁺	(i)	Group-III	(B)	As ³⁺	(ii)	Group-IIA	(C)	Cu ²⁺	(iii)	Group-IV	(D)	Al ³⁺	(iv)	Group-IIB	A
List-I (Metal ion)		List-II (Group in qualitative analysis)																				
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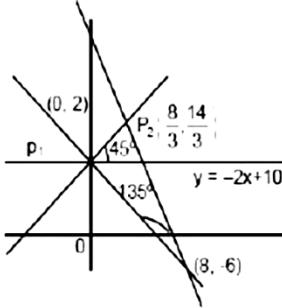
	Choose the most appropriate answer from the options given below.	
	(A) – (iii), (B) – (iv), (C) – (ii), (D) – (i)	
	(A) – (iv), (B) – (ii), (C) – (iii), (D) – (i)	
	(A) – (i), (B) – (iv), (C) – (ii), (D) – (iii)	
	(A) – (i), (B) – (ii), (C) – (iii), (D) – (iv)	
Sol.	Group-IIA : Hg^{2+} , Pb^{2+} , Cu^{2+} , Bi^{3+} , Cd^{2+} Group-IIB : As^{3+} , Sb^{3+} , Sn^{2+} ; Group-III : Al^{3+} , Cr^{3+} , Fe^{3+} Group-IV : Zn^{2+} , Mn^{2+} , Co^{2+} , Ni^{2+}	
45.	Thin layer chromatography of a mixture shows the following observation:  The correct order of elution in the silica gel column chromatography is	A
	A, C, B	
	B, C, A	
	C, A, B	
	B, A, C	
Sol.	According to the observation, A is more stable and interacts with the mobile phase more than C, and C is more drawn to the mobile phase than B. Hence, the correct order of elution in the silica gel column chromatography is: $B < C < A$.	
46.	25 mL of the given HCl solution requires 30 mL of 0.1 sodium carbonate solution. What is the volume of this solution required to titrate 30 mL of 0.2 M aqueous NaOH solution?	25
Sol.	For Na_2CO_3 (N_2) = $2 \times 0.1 \text{ N}$ $N_1 \times 25 = 2 \times 0.1 \times 30$; $N_1 = \frac{2 \times 0.1 \times 30}{25} = 0.24$ For titration with NaOH $N_1 V_1 = N_2 V_2 \Rightarrow 0.24 \times V_1 = 0.2 \times 30$ $V_1 = \frac{0.2 \times 30}{0.24} = 25 \text{ mL}$	
47.	A source of monochromatic radiation of wavelength 400 nm provides 1000 J of energy in 10 seconds. When this radiation falls on the surface of sodium, $x \times 10^{20}$ electrons are ejected per second. Assume that wavelength 400 nm is sufficient for ejection of electron from the surface of sodium metal. The value of x is _____. (Nearest interger) ($h = 6.626 \times 10^{-34} \text{ J s}$)	2

Sol.	According to the question, work function of metal is equal to energy of photon. $\lambda_{\text{photon}} = 400 \text{ nm}$ $E_{\text{photon}} = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{400 \times 10^{-9}} = 4.95 \times 10^{-19} \text{ J}$ In 10 seconds, the amount of energy provided is 1000 J. In 1 second, the amount of energy provided is 100 J. Number of photons emitted per second $= \frac{\text{total energy}}{\text{energy of one photon}}$ $\text{Number of photons} = \frac{100}{4.95 \times 10^{-19}} = 2.02 \times 10^{20} \text{ photons}$ So, 2.02×10^{20} photons eject 2.02×10^{20} electrons from the sodium metal.	
48.	The total number of isomers for a square planar complex $[\text{M}(\text{F})(\text{Cl})(\text{SCN})(\text{NO}_2)]$ is	12
49.	An athlete is given 100 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) for energy. This is equivalent to 1800 kJ of energy. The 50% of this energy gained is utilized by the athlete for sports activities at the event. In order to avoid storage of energy, the weight of extra water he would need to perspire is _____ g. (Nearest integer) Assume that there is no other way of consuming stored energy. Given : The enthalpy of evaporation of water is 45 kJ mol^{-1}. Molar mass of C, H & O are 12, 1 and 16 g mol^{-1}.	360
Sol.	$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$ Energy needed to perspire = $1800 \times \frac{1}{2} = 900 \text{ kJ}$ $900 \text{ kJ} = n_{\text{H}_2\text{O}} \times 45 \text{ kJ mol}^{-1}$ Moles of water needed to perspire $n_{\text{H}_2\text{O}} = \frac{900 \text{ kJ}}{45 \text{ kJ mol}^{-1}} = 20 \text{ mol}$ Mass of water needed to perspire = $20 \text{ mol} \times 18 \text{ g mol}^{-1} = 360 \text{ g}$	
50.	How many statements are correct (i) The pH of a mixture containing 400 mL of 0.1 M H_2SO_4 and 400 mL of 0.1 M NaOH will be approximately 1.3. (ii) Ionic product of water is temperature dependent. (iii) A monobasic acid with $K_a = 10^{-5}$ has a pH = 5. The degree of dissociation of this acid is 50%. (iv) The Le-Chatelier's principle is not applicable to common-ion effect.	3

Sol.	(i) For an acid-base mixture : $N_3V_3 = N_1V_1 - N_2V_2$ Normality = Molarity $\times$ basicity For H_2SO_4; $N_1 = 0.1 \times 2$, $V_1 = 400$ For $NaOH$; $N_2 = 0.1 \times 1$, $V_2 = 400$ $N_3V_3 = 0.2 \times 400 - 0.1 \times 400 = 40$ $N_3 \times 800 = 40 \Rightarrow N_3 = \frac{40}{800} = 0.05$ $pH = -\log[H^+] = -\log(0.05) = 1.3$ (ii) Ionic product of water is the product of molar concentration of H^+ ions and OH^- ions. Ionic product of water is temperature dependent. (iii) $K_a = 10^{-5}$; $pH = 5 \Rightarrow [H^+] = 10^{-5}$ $K_a = \frac{C\alpha^2}{1-\alpha} \Rightarrow K_a = \frac{[H^+]\alpha}{(1-\alpha)}$ $\Rightarrow 10^{-5} = \frac{10^{-5}\alpha}{1-\alpha} \Rightarrow 1-\alpha = \alpha$ $\alpha = \frac{1}{2} = 50\%$ (iv) Le-Chatelier's principle is applicable for common-ion effect.	
51.	If the circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2(3 - \sqrt{3})x + 2(4 - \sqrt{6})y = k + 6\sqrt{3} + 8\sqrt{6}$, $k > 0$ touch internally at the point $P(-\sqrt{3} - \lambda, -\mu - \sqrt{6})$, where $\lambda, \mu \in N$, then $2\lambda + \mu =$	A
	10	
	11	
	9	
	7	
	$C_1 = (-3, -4), r_1 = \sqrt{9+16-16} = 3$ $C_2 = (\sqrt{3}-3, \sqrt{6}-4), r_2 = \sqrt{(\sqrt{3}-3)^2 + (\sqrt{6}-4)^2 + k + 6\sqrt{3} + 8\sqrt{6}}$ $= \sqrt{34+k}$ Use $C_1C_2 = r_1 - r_2$ $3 = \sqrt{34+k} - 3$ $k = 2 \text{ and } r_2 = 6$ Point of contact P divides $\overline{C_1C_2}$ in the ratio $r_1:r_2 = 3:6$ externally	
52.	Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. Number of prime factors of $P^{-1}AP - 2I$ is equal to	A
	26	
	69	
	3	
	2	

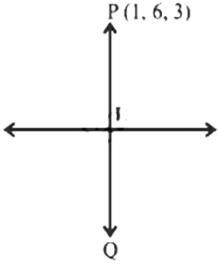
Sol.	$ P^{-1}AP - 2P^{-1}IP = P^{-1}(A - 2I)P $ $= P^{-1} A - 2I P = \frac{1}{ P } A - 2I P $ $= A - 2I = \begin{vmatrix} 0 & 1 & 2 \\ 6 & 0 & 11 \\ 3 & 3 & 0 \end{vmatrix} = 69 = 3 \times 23$ $2 + 23 = 26$	
53.	An ellipse and a hyperbola having centre at origin and foci on $(\pm 10, 0)$. If the distance between their directrices (both in positive direction of x axis) is 10 units, then positive difference of square of length of their major axis and transverse axis is _____.	A
	400	
	300	
	200	
	100	
Sol.	We have to find $(2a_1)^2 - (2a_2)^2$ $a_1e_1 = a_2e_2 = 10$ and $e_1 = \frac{10}{a_1} ; e_2 = \frac{10}{a_2}$ also $\left \frac{a_1}{e_1} - \frac{a_2}{e_2} \right = 10 ; x = \frac{a_2}{e_2} ; x = \frac{a_1}{e_1}$  $\left \frac{a_1^2}{10} - \frac{a_2^2}{10} \right = 10, a_1^2 - a_2^2 = 100$ $ 4a_1^2 - 4a_2^2 = 400$	
54.	If $f'(x) + (f'(x))^2 + (f'(x))^3 + (f'(x))^4 + \dots \infty = e^x$, where $f'(x) \in (-1, 1)$ and $f(0) = 0$ then the value of $\lim_{x \rightarrow 0} (1 + f(x))^{\frac{1}{x}}$ is equal to:	B
	e	
	$\sqrt{e}$	
	$1/\sqrt{e}$	
	1	

Sol.	$e^x = \frac{f'(x)}{1 - f'(x)}; f'(x) = \frac{e^x}{1 + e^x};$ $f(x) = \ln(1 + e^x) + C$ $f(0) = 0 \Rightarrow \ln(2) + C = 0 \Rightarrow C = -\ln 2$ $f(x) = \ln(1 + e^x) - \ln 2 \Rightarrow \lim_{x \rightarrow 0} (1 + f(x))^{1/x} = \sqrt{e}$	
55.	There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points $P_6, P_7, \dots, P_{11}$ on the side BC and 7 points $P_{12}, P_{13}, \dots, P_{18}$ on the side CA of the triangle. The number of triangles, that can be formed using the points $P_1, P_2, \dots, P_{18}$ as vertices, is:	B
	776	
	751	
	796	
	771	
Sol.	Number of ways selecting three points from side $AB = {}^5C_3$ Number of ways selecting three points from side $BC = {}^6C_3$ Number of ways selecting three points from side $AC = {}^7C_3$ Total number of triangle possible formed using the points $P_1 P_2 \dots P_{18}$ $= {}^{18}C_3 - {}^5C_3 - {}^6C_3 - {}^7C_3$ $= 816 - 10 - 20 - 35 = 751$	
56.	If the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the line $\frac{x}{7} + \frac{y}{2\sqrt{6}} = 1$ on the x-axis and the line $\frac{x}{7} - \frac{y}{2\sqrt{6}} = 1$ on the y-axis, then the eccentricity of the ellipse is	A
	$\frac{5}{7}$	
	$\frac{2\sqrt{6}}{7}$	
	$\frac{3}{7}$	
	$\frac{2\sqrt{5}}{7}$	

Sol.	Put $y = 0$ in $\frac{x}{7} + \frac{y}{2\sqrt{6}} = 1$, we get $x = 7$; So, ellipse intersect x-axis at 7, i.e., $a = 7$ Put $x = 0$ in $\frac{x}{7} - \frac{y}{2\sqrt{6}} = 1$, we get $y = -2\sqrt{6}$ so, ellipse intersect y-axis at $-2\sqrt{6}$ i.e., $b = -2\sqrt{6}$ $\therefore e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{24}{49}} = \frac{5}{7}$	
57.	A line L_1 ; $y = x + 2$ is rotated through an angle of 90° in counter-clockwise direction about the point $(0, 2)$ in xy plane and let new position of the line be $L_2 = 0$. Let L_3 ; $y = -2x + 10$. The area of the triangle formed by lines $L_1 = 0, L_2 = 0$ and $L_3 = 0$ is equal to:	C
	56/3	
	28/3	
	64/3	
	32/3	
Sol.	 L_2 is $y = -x + 2$ $\text{Area of triangle} = \frac{1}{2} \begin{vmatrix} 0 & 2 & 1 \\ 8 & 6 & 1 \\ 8/3 & 14/3 & 1 \end{vmatrix} = \frac{64}{3}$	
58.	Remainder when $2025^{2024^{2023}}$ is divided by 7 is	B
	1	
	4	
	3	
	2	
Sol.	$(2023 + 2)^n = 7k + 2^n$ $n = 2024^{2023} = (205 - 1)^{2023} = 2q - 1 = 3p + 2$ $7k + 2^n = 7k + 4(1 + 7)^p = 7m + 4$	
59.	Let $x, y \in R$ and $i = \sqrt{-1}$ satisfy the equation $(x + iy)^{2022} = x - iy$. If the number of possible distinct ordered pairs (x, y) satisfying the given equations is N then the value of product of non-zero digits of N is equal to:	C
	12	
	4	
	16	
	8	

Sol.	$\because z \cdot \bar{z} = z ^2$ $z^{2022} = \bar{z}$ $\Rightarrow z ^{2022} = \bar{z} = z $ $\Rightarrow z (z ^{2021} - 1) = 0$ $\Rightarrow (0, 0) \text{ and } z ^{2021} = 1$ $\Rightarrow z = 1$ $\therefore z^{2022} = \bar{z}$ $\Rightarrow z^{2023} = z\bar{z} = 1$ $\therefore z^{2023} = 1$ $\Rightarrow 2023 \text{ distinct solutions}$ $\therefore \text{So total (x, y) pairs are } N = 2024$	
60.	Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a} \cdot \vec{b} = 0$. For some $x, y \in R$, let $\vec{c} = x\vec{a} + y\vec{b} + (\vec{a} \times \vec{b})$. If $ \vec{c} = 2$ and the vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $x^2 + y^2 + 8 \cos^2 \alpha$ is equal to:	C
	8	
	4	
	3	
	6	
Sol.	We have, $\vec{c} = x\vec{a} + y\vec{b} + \vec{a} \times \vec{b} \quad \text{and} \quad \vec{a} \cdot \vec{b} = 0$ $ \vec{a} = \vec{b} = 1 \text{ and } \vec{c} = 2$ Also, given $\vec{c}$ is inclined on $\vec{a}$ and $\vec{b}$ with same angle α. $\therefore \vec{a} \cdot \vec{c} = x \vec{a} ^2 + y(\vec{a} \cdot \vec{b}) + \vec{a} \cdot (\vec{a} \times \vec{b})$ $ \vec{a} \vec{c} \cos \alpha = x + 0 + 0$ $x = 2 \cos \alpha$ Similarly, $ \vec{b} \vec{c} \cos \alpha = 0 + y + 0$ $\Rightarrow y = 2 \cos \alpha$ $ \vec{c} ^2 = x^2 + y^2 + \vec{a} \times \vec{b} ^2$ $4 = 8 \cos^2 \alpha + \vec{a} ^2 \vec{b} ^2 \sin^2 90^\circ$ $4 = 8 \cos^2 \alpha + 1 \Rightarrow 8 \cos^2 \alpha = 3$	
61.	The area of the largest rectangle having two vertices lying on the x-axis and the remaining vertices lying above the x-axis and on the parabola $y = 9 - x^2$ is equal to:	A
	$12\sqrt{3}$ Sq. units	
	$6\sqrt{3}$ Sq. units	
	$24\sqrt{3}$ Sq. units	
	$5\sqrt{3}$ Sq. units	

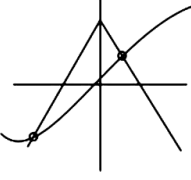
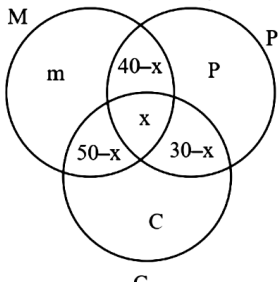
Sol.	For reasons of symmetry, the rectangle will be symmetric with respect to the y axis and thus the two vertices lying on the x axis have coordinates $(-x, 0)$ and $(x, 0)$ and the ones lying on the parabola have coordinates $(-x, 9 - x^2)$ and $(x, 9 - x^2)$, and because they must lie above the x axis we can consider only $x \in [0, 3)$. We have to maximize the area: $A(x) = 2x(9 - x^2) \text{ for } x \in [0, 3)$ $A(x) = 18x - 2x^3$ $\frac{dA}{dx} = 18 - 6x^2 = 6(3 - x^2)$ To find the critical points we equate the derivative to zero: $\frac{dA}{dx} = 0 \Rightarrow x = \sqrt{3}$ and choose the solution in the interval $[0, 3)$ $\frac{d^2A}{dx^2} = -9x$ As for $x = \sqrt{3}$, we have that $\frac{d^2A}{dx^2} < 0$, the critical point is a maximum and the the maximum area is: $A(\sqrt{3}) = 2\sqrt{3}(9 - 3) = 12\sqrt{3}$	
62.	Let $y = y(x)$ be the solution of the differential equation $(\operatorname{cosec} x)dy + (2(1 - x)\cot x + x(x - 2))dx = 0$ such that $y\left(\frac{\pi}{2}\right) = 3$. The value of $y(2)$ is equal to:	D
	2	
	$2(1 - \cos 2)$	
	$2(1 + \cos 2)$	
	3	
Sol.	$dy = d((x^2 - 2x)\cos x)$ $\therefore y(x) = (x^2 - 2x)\cos x + 3$	
63.	If (α, β, γ) is the image of the point $(1, 6, 3)$ with respect to the line $\frac{x}{2} = \frac{y-1}{4} = \frac{z-2}{6}$ then the value of $\alpha^2 + \beta^2 + \gamma^2$ is equal to:	D
	26	
	37	
	17	
	50	

Sol.	Let P(1, 6, 3) be the given point and let L be the foot of perpendicular from P to the given line.  The coordinates of a general point on the given line are $\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-2}{3} = \lambda$ i.e., $x = \lambda, y = 2\lambda + 1, z = 3\lambda + 2$. If the coordinates of L are $(\lambda, 2\lambda + 1, 3\lambda + 2)$, then the direction ratios of PL are $\lambda - 1, 2\lambda - 5, 3\lambda - 1$. But the direction ratios of given line which is perpendicular to PL are 1, 2, 3. Therefore, $(\lambda - 1)1 + (2\lambda - 5)2 + (3\lambda - 1)3 = 0$, which gives $\lambda = 1$. Hence coordinates of L are (1, 3, 5). Let Q(x_1, y_1, z_1) be the image of P(1, 6, 3) in the given line. Then L is the mid-point of PQ. Therefore, $\frac{x_1 + 1}{2} = 1$ $\frac{y_1 + 6}{2} = 3$ $\frac{z_1 + 3}{2} = 5$ $\Rightarrow x_1 = 1, y_1 = 0, z_1 = 7$ Hence, the image of (1, 6, 3) in the given line is (1, 0, 7).	
64.	Two integers x and y are chosen one after the other with replacement from the set {0, 1, 2, 3,, 20}. Then the probability that $x - y > 8$ is equal to:	A
	$\frac{52}{147}$	
	$\frac{400}{441}$	
	$\frac{361}{441}$	
	$\frac{77}{147}$	

Sol.	When $y - x > 8$, the following choices can be made: Suppose $x = 0$, then y can be any one of 9, 10, 11, ..., 20 $\rightarrow$ 12 ways Similarly $x = 1 \rightarrow y$ can be 10, 11, ..., 20 $\rightarrow$ 11 ways $x = 1 \rightarrow y$ can be 11, 12, ..., 20 $\rightarrow$ 10 ways $\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$ $x = 11 \Rightarrow y = 20 \rightarrow 1$ way $\therefore$ the total number of ways = $1 + 2 + \dots + 12 = 6 \times 13 = 78$ Similarly the number of ways for $x - y > 8$ is 78 Total number of ways for $x - y > 8$ is 78 . Total number of all possible ways = 21×21 Hence the required probability = $\frac{2 \times 78}{21 \times 21} = \frac{52}{147}$													
65.	Let $f(x) = \left[\log(\sin^{-1}(\sqrt{x^2 + 3x + 2})) \right]$ where $[.]$ is G.I.F. Which of the following is true?	C												
	Domain of defintion $f(x) = \left[\frac{-3-\sqrt{5}}{2}, \frac{-3+\sqrt{5}}{2} \right]$.													
	Domain of definition of $f(x)$ is $\left[\frac{-3+\sqrt{5}}{3}, -1 \right)$.													
	Range of $f(x)$ is the set of all non-positive integers.													
	Range of $f(x)$ is the set of all positive integers.													
Sol.	$x^2 + 3x + 2 > 0$ $\Rightarrow x \in (-\infty, -2) \cup (-1, \infty)$ $x^2 + 3x + 2 \leq 1$ $\Rightarrow x \in \left[\frac{-3-\sqrt{5}}{2}, \frac{-3+\sqrt{5}}{2} \right]$ $x \in \left[\frac{-3-\sqrt{5}}{2}, -2 \right) \cup \left(-1, \frac{-3+\sqrt{5}}{2} \right]$ $0 < \sin^{-1}(\sqrt{x^2 + 3x + 2}) \leq \frac{\pi}{2}$ $-\infty < \log(\sin^{-1}(\sqrt{x^2 + 3x + 2})) \leq \log \frac{\pi}{2}$ $\therefore$ set of all non positive integers.													
66.	<table><tr><td>Class interval:</td><td>0-6</td><td>6-12</td><td>12-18</td><td>18-24</td><td>24-30</td></tr><tr><td>Frequency</td><td>4</td><td>5</td><td>3</td><td>6</td><td>2</td></tr></table> The mean deviation from the median of the given data is equal to:	Class interval:	0-6	6-12	12-18	18-24	24-30	Frequency	4	5	3	6	2	A
Class interval:	0-6	6-12	12-18	18-24	24-30									
Frequency	4	5	3	6	2									
	7													
	7.5													
	6													
	6.5													

Sol.	<table><tr><th>Class-interval</th><th>f_i</th><th>x_i</th><th>c.f.</th><th>$d_i = x_i - \text{Med}$</th><th>$f_i d_i$</th></tr><tr><td>0-6</td><td>4</td><td>3</td><td>4</td><td>11</td><td>44</td></tr><tr><td>6-12</td><td>5</td><td>9</td><td>9</td><td>5</td><td>25</td></tr><tr><td>12-18</td><td>3</td><td>15</td><td>12</td><td>1</td><td>3</td></tr><tr><td>18-24</td><td>6</td><td>21</td><td>18</td><td>7</td><td>42</td></tr><tr><td>24-30</td><td>2</td><td>27</td><td>20</td><td>13</td><td>26</td></tr><tr><td></td><td>N = 20</td><td></td><td></td><td></td><td>$\sum f_i d_i = 140$</td></tr></table>	Class-interval	f_i	x_i	c.f.	$d_i = x_i - \text{Med} $	$f_i d_i$	0-6	4	3	4	11	44	6-12	5	9	9	5	25	12-18	3	15	12	1	3	18-24	6	21	18	7	42	24-30	2	27	20	13	26		N = 20				$\sum f_i d_i = 140$	
	Class-interval	f_i	x_i	c.f.	$d_i = x_i - \text{Med} $	$f_i d_i$																																						
	0-6	4	3	4	11	44																																						
	6-12	5	9	9	5	25																																						
	12-18	3	15	12	1	3																																						
	18-24	6	21	18	7	42																																						
	24-30	2	27	20	13	26																																						
		N = 20				$\sum f_i d_i = 140$																																						
Median class = $\left(\frac{N}{2}\right)$ th term = $\frac{20}{2}$ th term = 10th term i.e. 12-18																																												
$\therefore$ Median = $l + \frac{N/2 - cf}{f} \times h$																																												
$= 12 + \frac{10 - 9}{3} \times 6 = 12 + \frac{1}{3} \times 6 = 12 + 2 = 14$																																												
and MD = $\frac{\sum f_i d_i}{N} = \frac{140}{20} = 7$																																												
Hence, the required MD = 7																																												
67.	Let $A = \begin{bmatrix} 1 + \cos^2 \theta & \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & 1 + \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & \sin^2 \theta & 1 + 4 \cos 6\theta \end{bmatrix}$ and $f(\theta) = \det A$. The minimum value of $f(\theta)$ for $\theta \in \left[\frac{\pi}{18}, \frac{\pi}{9}\right]$ is equal to:	B																																										
	-2																																											
	0																																											
	4																																											
	$2 - 2\sqrt{3}$																																											
Sol.	$f(\theta) = 2 + 4 \cos \theta$ $6\theta \in \left[\frac{\pi}{3}, \frac{2\pi}{3}\right]$ $\therefore \max = f\left(\frac{2\pi}{3}\right) = 0$																																											
68.	Let $f: R \rightarrow R$ be a function defined as $f(x) = \begin{cases} 3\left(1 - \frac{ x }{2}\right), & \text{if } x \leq 2 \\ 0, & \text{if } x > 2 \end{cases}$. Let $g: R \rightarrow R$ be given by $g(x) = f(x + 2) - f(x - 2)$. If n and m denote the number of points in R, where g is not continuous and not differentiable, respectively, then $n + m$ is equal to _____.	A																																										
	4																																											
	16																																											
	34																																											
	48																																											
Sol.																																												
69.	Let $f(x)$ be a polynomial of degree 6 in x, in which the coefficient of x^6 is unity and it has extrema at $x = -1$ and $x = 1$. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^3} = 1$, then $5 \cdot f(2)$ is equal to _____.	B																																										
	266																																											
	144																																											
	456																																											

	570	
Sol.		
70.	Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i} - \hat{j} + \hat{k}, \hat{i} + 2\hat{j} + \mu\hat{k}$ and $3\hat{i} - 4\hat{j} + 5\hat{k}$, where $\lambda - \mu = 5$, are coplanar, then $\sum_{(\lambda, \mu) \in S} 80(\lambda^2 + \mu^2)$ is equal to	A
	2290	
	2210	
	2370	
	2130	
Sol.	Since, the given vectors are coplanar. $\therefore \begin{vmatrix} \lambda & -1 & 1 \\ 1 & 2 & \mu \\ 3 & -4 & 5 \end{vmatrix} = 0$ $\Rightarrow \lambda(10 + 4\mu) + 1(5 - 3\mu) + 1(-4 - 6) = 0$ $\Rightarrow 10\lambda + 4\lambda\mu + 5 - 3\mu - 10 = 0$ $\Rightarrow 10(5 + \mu) + 4\mu(5 + \mu) - 5 - 3\mu = 0$ $\quad [\because \lambda - \mu = 5]$ $\Rightarrow 50 + 10\mu + 20\mu + 4\mu^2 - 5 - 3\mu = 0$ $\Rightarrow 4\mu^2 + 27\mu + 45 = 0$ $\Rightarrow 4\mu^2 + 12\mu + 15\mu + 45 = 0$ $\Rightarrow (4\mu + 15)(\mu + 3) = 0$ $\Rightarrow \mu = -3 \text{ and } \mu = \frac{-15}{4}$ $\Rightarrow \lambda = 2 \text{ and } \lambda = \frac{5}{4}$ $\therefore \sum_{(\lambda, \mu) \in S} 80(\lambda^2 + \mu^2)$ $= 80 \left[(4 + 9) + \left(\frac{25}{16} + \frac{225}{16} \right) \right] = \frac{458}{16} \times 80$ $= 2290$	
71.	If $[a, b]$ is the range of the function $f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$, then the number of roots of the equation $1 - x = \tan^{-1} x$ is $([b] + k)$. Value of k is . (where $[.]$ is G.I.F.)	0

Sol.	Domain of $\sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ is $[-1, 1]$ So range is $\left[\frac{\pi}{2} - \frac{\pi}{4}, \frac{\pi}{2} + \frac{\pi}{4}\right] = \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ $b = \frac{3\pi}{4} \Rightarrow [b] = 2$  Now no. of solutions of $1 - x = \tan^{-1}x$ 2 solutions.	
72.	In a survey of 220 students of a higher secondary school, it was found that at least 125 and at most 130 students studied Mathematics; at least 85 and at most 95 studied Physics; at least 75 and at most 90 studied Chemistry; 30 studied both Physics and Chemistry; 50 studied both Chemistry and Mathematics; 40 studied both Mathematics and Physics and 10 studied none of these subjects. Let m and n respectively be the least and the maximum number of students who studied all the three subjects. The value of m + n is equal to	45
Sol.	Let M: mathematics; P: Physics; C: Chemistry  $125 \leq m + 90 - x \leq 130 \dots (i)$ $85 \leq P + 70 - x \leq 95 \dots (ii)$ $75 \leq C + 80 - x \leq 90 \dots (iii)$ Also, $m + P + C + 40 - x + 30 - x + 50 - x + x = 210$ $\Rightarrow m + P + C + 120 - 2x = 210$ $\Rightarrow m + P + C - 2x = 90$ $\Rightarrow 15 \leq x \leq 45 \text{ \& } 30 - x \geq 0$ $\Rightarrow 15 \leq x \leq 30$ So, m = 15 and n = 30	
73.	The value of $\int_{-1}^2 \left[\frac{50x^2 + [x] + 50}{1 + x^2} \right] dx$ (where $[.]$ denotes greatest integer function) is equal to	149

Sol.	$[x] = 0, \forall x \in [0, 1)$ For $x \in [1, 2), [x] = 1$ $\Rightarrow \frac{[x]}{1+x^2} = \frac{1}{1+x^2} < 1, \forall x \in [1, 2) \Rightarrow \left[\frac{[x]}{1+x^2} \right] = 0$ For $x \in [-1, 0), [x] = -1 \Rightarrow \frac{[x]}{1+x^2} = -\frac{1}{1+x^2}$ Clearly, $2 \geq 1+x^2 > 1, \forall x \in [-1, 0)$ $\frac{1}{2} \leq \frac{1}{1+x^2} < 1 \Rightarrow -\frac{1}{2} \geq -\frac{1}{1+x^2} > -1$ $\Rightarrow \left[\frac{[x]}{1+x^2} \right] = -1 \forall x \in [-1, 0)$ $\therefore$ Given integral $= \int_{-1}^2 50 dx + \int_{-1}^0 -1 dx + \int_0^2 0 dx = 150 - 1 = 149$	
74.	Let $a = 5^2 + 11^2 + 19^2 + 29^2 + 41^2 + \dots + 10 \text{ terms}$ and $b = \sum_{n=1}^{10} n^4$. The value of $\left[\frac{a-b}{909} \right]$ (where $[.]$ denotes G.I.F.) is equal to	25
Sol.	$a = \sum_{n=1}^{10} (n^2 + 3n + 1)^2 = \sum_{n=1}^{10} (n^4 + 6n^3 + 11n^2 + 6n + 1)$ $\Rightarrow a - b = \sum_{n=1}^{10} (6n^3 + 11n^2 + 6n + 1) = 22725$	
75.	Three points $O(0, 0), P(a, a^2), Q(-b, b^2), a > 0, b > 0$, are on the parabola $y = x^2$. Let S_1 be the area of the region bounded by the line PQ and the parabola, and S_2 be the area of the triangle OPQ. If the minimum value of $\frac{S_1}{S_2}$ is $\frac{m}{n}$. $\gcd(m, n) = 1$, then $m + n$ is equal to _____.	7
Sol.	(7) : Equation of line PQ is, $y - a^2 = \frac{b^2 - a^2}{-b - a}(x - a)$ $\Rightarrow y = x(a - b) + ab$ $\therefore S_1 = \int_{-b}^a (x(a - b) + ab - x^2) dx$ $= \left[(a - b) \frac{x^2}{2} + abx - \frac{x^3}{3} \right]_{-b}^a$ $= (a - b) \frac{(a^2 - b^2)}{2} + ab(a + b) - \frac{(a^3 + b^3)}{3}$ $\Rightarrow S_1 = \frac{1}{6}(a + b)^3$ Also, $S_2 = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ a & a^2 & 1 \\ -b & b^2 & 1 \end{vmatrix} = \frac{1}{2} ab(a + b)$ Now, $\frac{S_1}{S_2} = \frac{(a + b)^3 / 6}{ab(a + b) / 2} = \frac{1}{3} \frac{(a + b)^2}{ab}$ $= \frac{a^2 + b^2 + 2ab}{3ab} = \frac{a}{3b} + \frac{b}{3a} + \frac{2}{3}$ $= \frac{1}{3} \left(\frac{a}{b} + \frac{b}{a} + 2 \right)$ Now, $\frac{a}{b} + \frac{1}{a/b} \geq 2$ $\therefore$ Minimum value of $\frac{S_1}{S_2} = \frac{4}{3}$ So, $m + n = 4 + 3 = 7$	